



SECI1143

PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 03 GROUP 06

PSDA ASSIGNMENT 1

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Assignment 1.

- 1a) Population is the entire collection of individuals or objects while sample is the subset of the population.

In this situation, the population represents all students in Faculty of Computing in the university. The sample is the representative selected subgroup of students from the Faculty of Computing.

- bi) discrete / continuous

discrete: total number of students / number of courses taken in a semester / number of siblings

continuous: students' CGPA / family income per year

- ii) categorical / numerical

categorical: ethnicity / type of property

numerical: students' CGPA / family income per year / number of siblings.

- iii) level of measurement.

nominal: ethnicity / type of property

ordinal: total expenses throughout the year

interval: number of courses taken / number of each semester.

ratio: family income

- c) Primary Data Source.

Students' socioeconomic backgrounds and academic performance.

- d) Secondary Data Source.

University records of students' database, academic transcripts & financial records

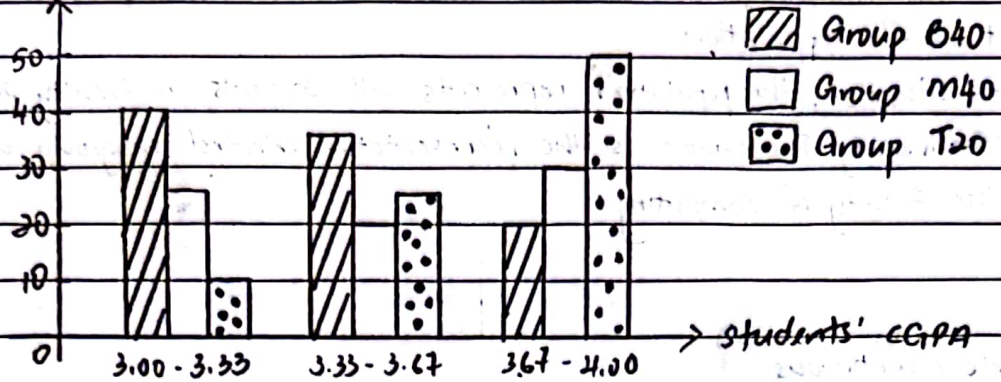
- ~~e) mode for ethnicity: Chinese.~~

~~mean GPA: 3.97~~

~~median of the yearly family income: RM 80,000~~

~~range of the yearly family income: RM 35,000~~

e) no. of students



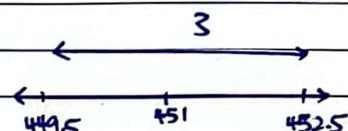
2)

	Frequency	Relative Frequency
0	3	$\frac{3}{30} = 0.10$
1	2	$\frac{2}{30} = 0.07$
2	7	$\frac{7}{30} = 0.23$
3	7	$\frac{7}{30} = 0.23$
4	5	$\frac{5}{30} = 0.17$
5	4	$\frac{4}{30} = 0.13$
6	2	$\frac{2}{30} = 0.07$
Total	30	1.00

3a) weight	Tabulation	Frequency	weight	Tabulation	Frequency
450		10	462		2
451		2	463		3
452		2	464		4
453		7	465		3
454		3	466		4
455		6	467		5
456		4	468		0
457		6	469		4
458		4	470		5
459		3	471		8
460		3	472		9
461		2	473		1

a) Histogram

$$\begin{aligned}
 \text{Range} &= X_h - X_l \\
 &= 473 - 450 \\
 &= 23
 \end{aligned}$$



cell interval

Cell midpoint

$$i = \frac{R}{1 + 3.322 \log n}$$

$$MP_i = X_l + \frac{i}{2} = 450 + \frac{3}{2} \approx 451$$

$$i = \frac{23}{1 + 3.322 \log 100}$$

$$i = 3.01$$

trial and error

$$i = 2 ; h = \frac{23}{2} = 12$$

$$i = 3 ; h = \frac{23}{3} = 8$$

$$i = 4 ; h = \frac{23}{4} = 6$$

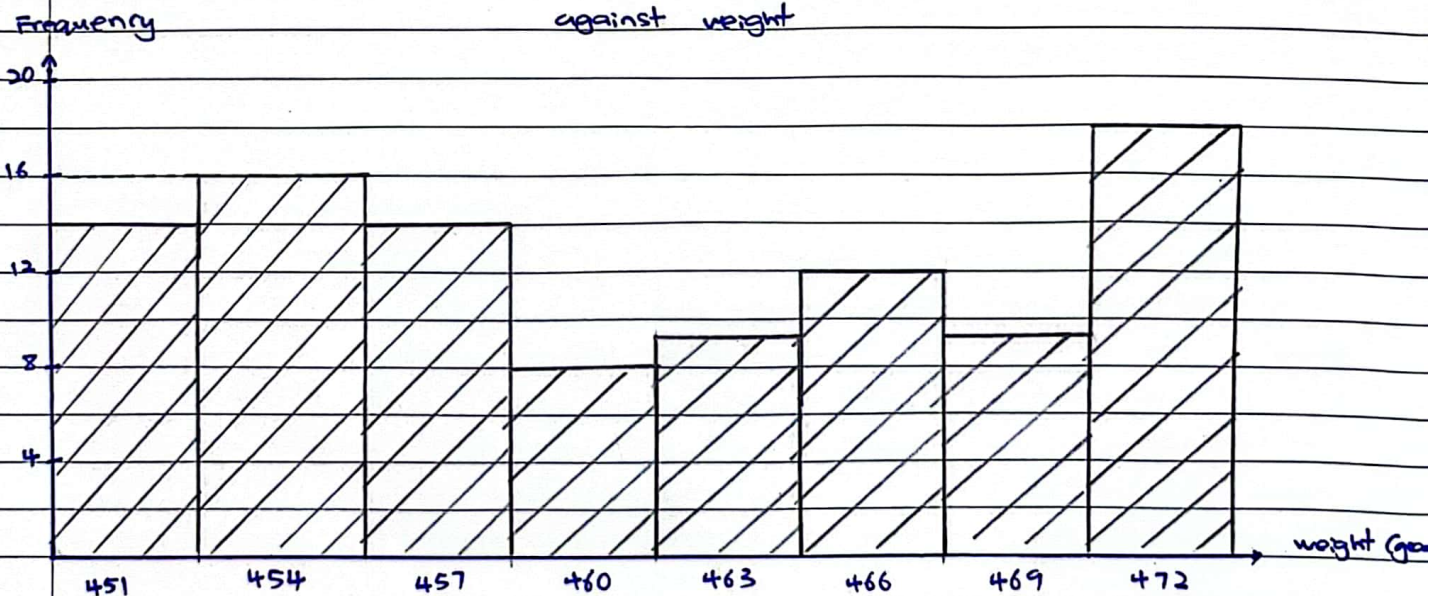
Cell midpoint
451
454
457
460
463
466
469
472

$\therefore i = 3$, give best presentation of data

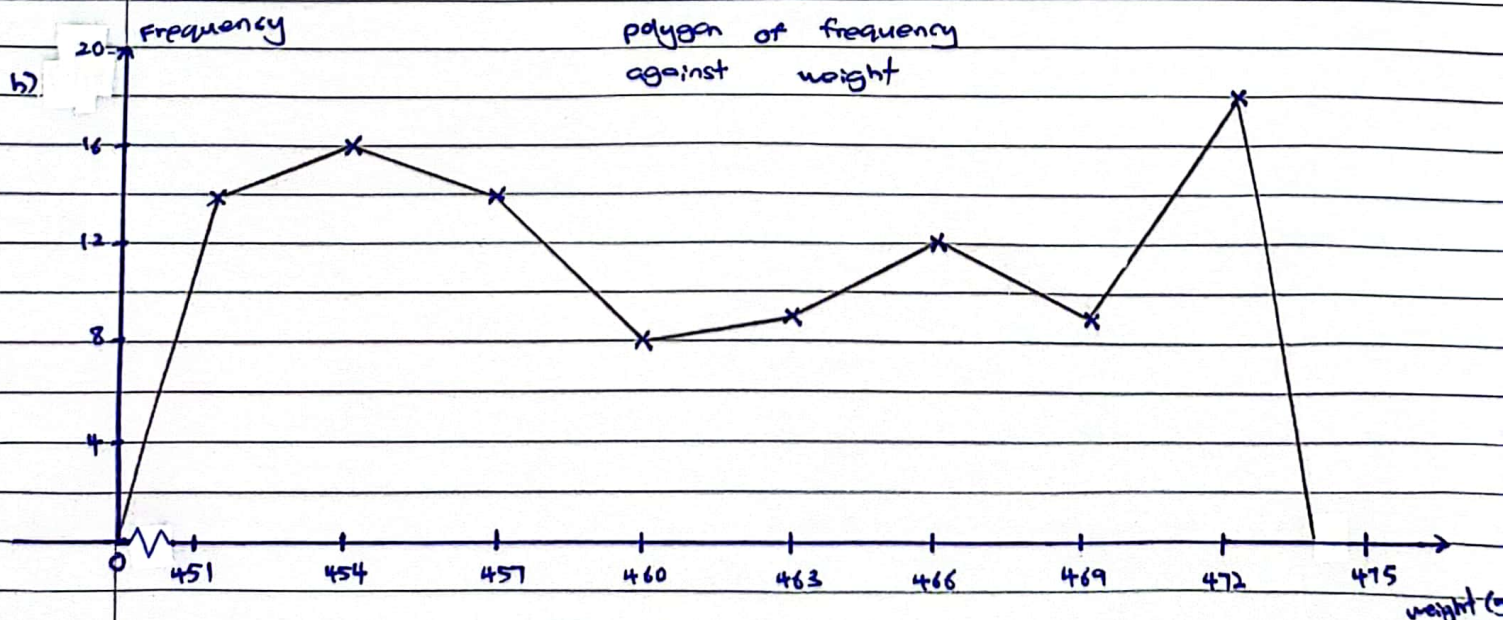


Cell Boundaries	Cell Midpoint	Frequency
449.5 - 452.5	451	14
452.5 - 455.5	454	16
455.5 - 458.5	457	14
458.5 - 461.5	460	8
461.5 - 464.5	463	9
464.5 - 467.5	466	12
467.5 - 470.5	469	9
470.5 - 473.5	472	18

Histogram of frequency
against weight



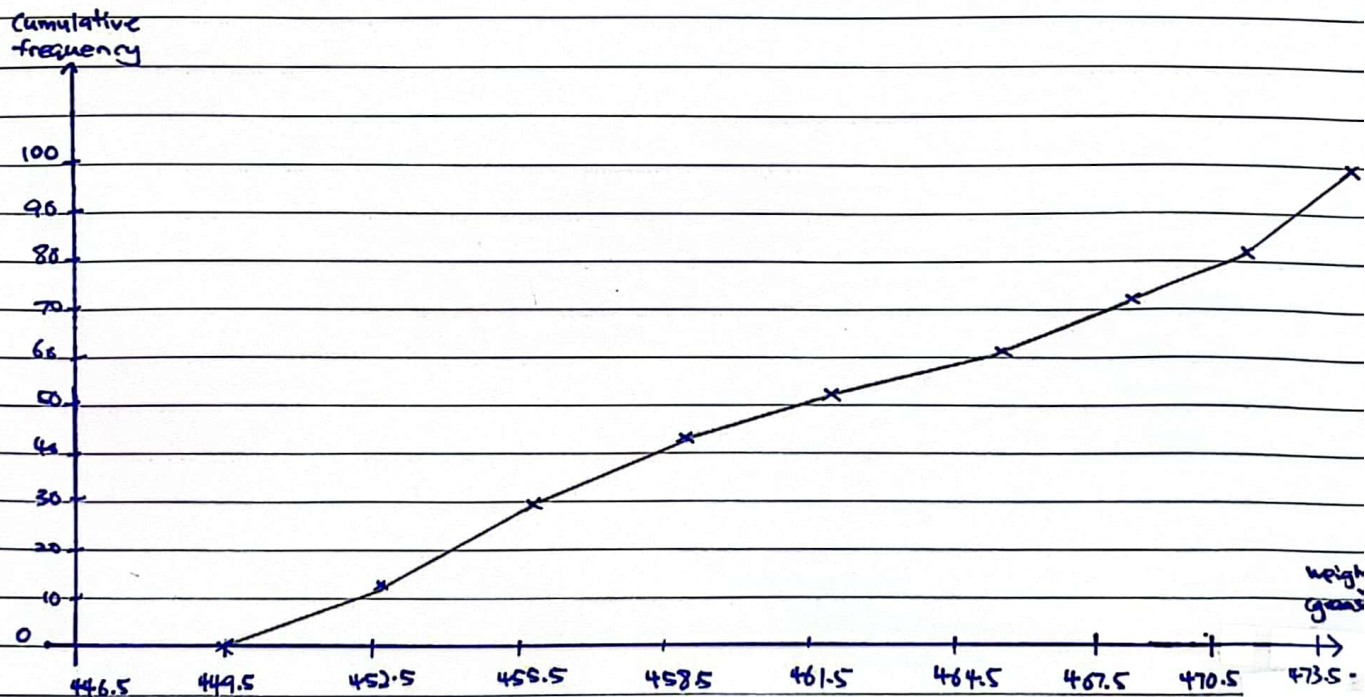
polygon of frequency
against weight



c)

cell boundaries	frequency	cumulative frequency
449.5 - 452.5	14	14
452.5 - 455.5	16	30
455.5 - 458.5	14	44
458.5 - 461.5	8	52
461.5 - 464.5	9	61
464.5 - 467.5	12	73
467.5 - 470.5	9	82
470.5 - 473.5	18	100

Ogive of cumulative frequency
against weight (grams)



Question 4

18	702	765	811	832	855	896
902	905	918	919	920	923	929
936	938	948	950	956	958	958
970	972	978	1009	1009	1022	1035
1037	1045	1067	1085	1092	1102	1122
1126	1151	1156	1157	1157	1162	1170
1195	1195	1196	1217	1237	1311	1333
1340						

N=50

$$Q_1: P=25, i = \frac{P}{100} \times N$$

$$i = \frac{25}{100} \times 50 = 12.5$$

$$k=13$$

$$Y[13] = 923$$

$$Q_2: P=50, i = \frac{P}{100} \times N$$

$$i = \frac{50}{100} \times 50 = 25$$

$$k=25$$

$$(Y[25] + Y[25+1]) \div 2$$

$$= (1009 + 1009) \div 2$$

$$= 1009$$

$$Q_3: P=75, i = \frac{P}{100} \times N$$

$$i = \frac{75}{100} \times 50 = 37.5$$

$$k=38$$

$$Y[38] = 1156$$

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

$$= 1156 - 923$$

$$= 233$$

$$\text{Lower Limit} : Q_1 - 1.5 \times \text{IQR} = 923 - 1.5(233) = 573.5$$

$$\text{Upper Limit} : Q_3 + 1.5 \times \text{IQR} = 1156 + 1.5(233) = 1505.5$$

Since 78 is not within the 25th percentile or Q_1 value - $1.5 \times \text{IQR}$ limit, it is considered an outlier.

